

$$\begin{aligned}
& -\frac{1}{16\pi^2}\left(-\frac{1}{3}g_2^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{2,1,0}[m_1,m_d^r]+\frac{2}{3}g_2^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{3,1,-1}[m_1,m_d^r]\right. \\
& +\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{4,1,-2}[m_1,m_d^r]-\frac{1}{36}g_1^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{2,1,0}[m_1,m_q^p]+\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{3,1,-1}[m_1,m_q^p] \\
& -\frac{1}{36}g_2^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{4,1,-2}[m_1,m_q^p]+\text{y}_u^{\text{112}}(g_1^2+3g_2^2)\text{LF}_{2,2,-1}[m_1,m_q^{\text{11}}]-\frac{1}{27}g_1^4\text{y}_u^{\text{112}}\text{LF}_{2,2,-1}[m_1,m_q^{\text{12}}]- \\
& \frac{1}{27}\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{2,1,0}[m_2,m_q^p]+\frac{2}{3}g_2^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{3,1,-1}[m_2,m_q^p]- \\
& \frac{1}{36}g_2^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{2,1,-2}[m_2,m_q^p]-\frac{1}{32}g_2^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}(g_1^2-g_2^2)\text{LF}_{2,2,-1}[m_2,m_q^{\text{11}}]- \\
& \frac{1}{32}\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{2,2,0}[m_3,m_q^p]+\frac{8}{32}g_2^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{3,1,-1}[m_3,m_q^p]- \\
& \frac{4}{32}g_2^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{2,1,0}[m_3,m_q^p]+\frac{4}{32}g_2^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{4,1,-2}[m_3,m_q^p]+\frac{4}{32}g_2^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{4,1,-2}[m_3,m_q^p] \\
& +\text{y}_u^{\text{112}}(g_1^2+3g_2^2)\text{LF}_{2,2,0}[m_3,m_q^p]-\frac{2}{12}g_1^2g_2^2\text{y}_u^{\text{112}}\text{LF}_{2,2,-1}[m_3,m_q^{\text{12}}]+\frac{1}{12}\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{2,1,0}[m_d^r,\tilde{\mu}]+ \\
& \frac{1}{4}g_1^2\tilde{\mu}^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{2,2,-1}[m_d^r,m_q^p]-\frac{1}{4}g_1^2\tilde{\mu}^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{2,2,0}[m_e^r,m_q^p]-\frac{1}{4}g_1^2\tilde{\mu}^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{3,2,-1}[m_q^p,m_e^r]+ \\
& -\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}(3g_1^2-g_2^2)\text{LF}_{4,1,-1}[m_q^p,m_e^p]+\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}(g_1^2-g_2^2)\text{LF}_{5,1,-2}[m_q^p,m_e^r]- \\
& \frac{1}{2}g_2^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{3,1,0}[m_q^p,m_d^r]-\frac{1}{4}g_1^2\tilde{\mu}^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{3,2,-1}[m_q^p,m_d^r]+\frac{1}{4}\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}(5g_1^2-3g_2^2)\text{LF}_{4,1,-1}[m_q^p,m_d^r]+ \\
& \overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}(-g_1^p+g_2^2)\text{LF}_{5,1,-2}[m_q^p,m_d^r]+\frac{1}{12}g_1^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{2,1,0}[m_q^p,\tilde{\mu}]+\frac{1}{432}g_1^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{2,1,0}[m_q^p,\tilde{\mu}]+ \\
& \frac{1}{432}g_1^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}(g_1^2+3g_2^2)\text{LF}_{2,2,-1}[m_q^p,\tilde{\mu}]+\frac{1}{432}g_1^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}(g_1^2+3g_2^2)\text{LF}_{2,2,-1}[m_q^p,\tilde{\mu}]+ \\
& \frac{1}{432}\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}(-7g_1^2+3g_2^2)\text{LF}_{3,1,0}[m_q^p,m_q^p]-\frac{1}{8}\text{y}_u^{\text{112}}\overline{y}^{\text{DP}^2}\text{a}_0^{\text{P}^2}\text{a}_0^{\text{P}^2}(g_1^2+3g_2^2) \\
& +\frac{3}{4}\text{y}_u^{\text{112}}\overline{y}^{\text{DP}^2}\text{a}_0^{\text{P}^2}\text{a}_0^{\text{P}^2}(5g_1^2-3g_2^2)\text{LF}_{4,1,-1}[m_u^r,m_q^p]+\frac{1}{6}g_1^2\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{LF}_{2,1,0}[m_u^r,\tilde{\mu}]- \\
& \frac{1}{6}\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{y}_u^{\text{1L}}\text{LF}_{2,2,-1}[m_u^r,\tilde{\mu}]-\frac{3}{2}\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{y}_u^{\text{1L}}\text{LF}_{2,2,0}[m_u^r,\tilde{\mu}]+\frac{4}{27}g_1^4\text{y}_u^{\text{112}}\text{LF}_{2,1,0}[m_q^{\text{12}},m_1]+ \\
& g_3^2\text{y}_u^{\text{112}}\text{LF}_{2,1,0}[m_u^{\text{12}},\tilde{\mu}]+\frac{1}{2}g_1^2\text{y}_u^{\text{112}}\text{LF}_{2,2,-1}(g_1^2-g_2^2)\text{LF}_{3,1,-1}[\tilde{\mu},m_1]+ \\
& \text{y}_u^{\text{112}}\text{LF}_{3,2,-2,0}[\tilde{\mu},m_1]+\frac{3}{8}g_1^2\text{y}_u^{\text{112}}\text{LF}_{2,2,-1}[\tilde{\mu},m_1]+\frac{1}{4}g_1^4\text{y}_u^{\text{112}}\text{LF}_{4,2,-3}[\tilde{\mu},m_1]- \\
& m_1^2\text{y}_u^{\text{112}}\text{LF}_{4,2,-2}[\tilde{\mu},m_1]+\frac{1}{2}g_1^2\text{y}_u^{\text{112}}(g_1^2-g_2^2)\text{LF}_{5,1,-3}[\tilde{\mu},m_1]+\text{y}_u^{\text{112}}(g_1^2-g_2^2)\text{LF}_{3,1,-1}[\tilde{\mu},m_2]+\frac{15}{8}g_1^4\text{y}_u^{\text{112}}\text{LF}_{3,2,-2}[\tilde{\mu},m_2]+ \\
& m_2^2\text{y}_u^{\text{112}}\text{LF}_{3,2,-1}[\tilde{\mu},m_2]+3g_2^2\text{y}_u^{\text{112}}(-g_1^2+g_2^2)\text{LF}_{4,1,-2}[\tilde{\mu},m_2]-\text{y}_u^{\text{112}}\text{LF}_{4,2,-3}[\tilde{\mu},m_2]-\frac{1}{2}g_1^2\text{y}_u^{\text{112}}(m_2^2+2\tilde{\mu}^2)\text{LF}_{4,2,-2}[\tilde{\mu},m_2]+ \\
& \text{y}_u^{\text{112}}\tilde{\mu}^2(g_1^2-g_2^2)\text{LF}_{5,1,-3}[\tilde{\mu},m_2]-\frac{1}{2}\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{y}_u^{\text{1L}}\text{LF}_{2,1,0}[\tilde{\mu},m_d^t]+\frac{1}{2}\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{y}_u^{\text{1L}}\text{LF}_{3,1,-1}[\tilde{\mu},m_d^t]- \\
& \overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{y}_u^{\text{1L}}\text{LF}_{4,1,-2}[\tilde{\mu},m_d^t]+\frac{1}{2}\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{y}_u^{\text{1L}}\text{LF}_{2,1,0}[\tilde{\mu},m_u^t]+\frac{1}{2}\overline{y}^{\text{DP}^2}\text{y}_d^{\text{1L}}\text{y}_u^{\text{P12}}\text{y}_u^{\text{1L}}\text{LF}_{4,1,-2}[\tilde{\mu},m$$